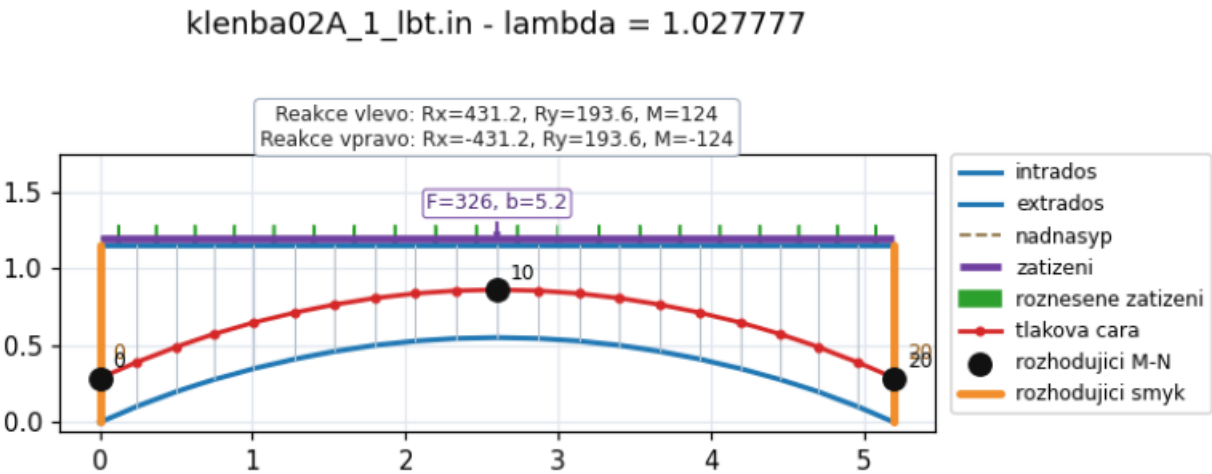


ArchLBT vypocetni protokol

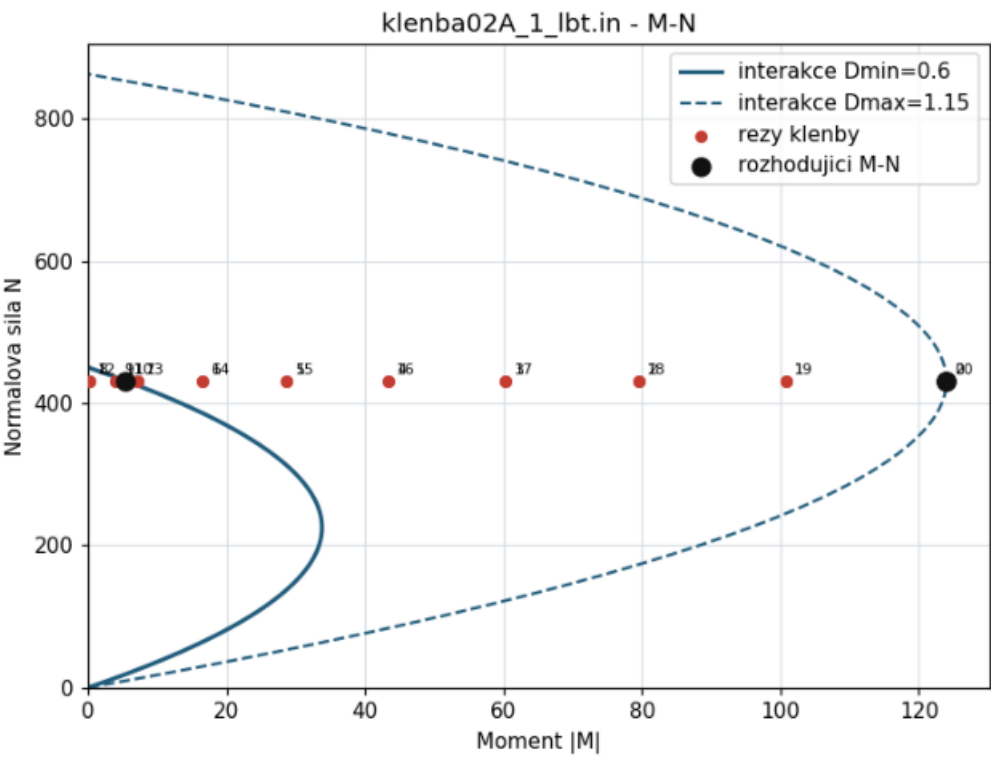
1) Zadani ulohy

```
# ArchLBT input
# Unit convention: use one consistent force-length system.
# Geometry is in length units, typical m. Unit weights are force/volume, typical kN/m3.
# Surface point_load force F is the total force on the loaded length, typical kN.
# Self-weight and fill loads are multiplied internally by arch_width; point_load F is not.
span = 5.2
rise = 0.55
geom_code = 1
# extrados_mode: normal = constant normal thickness, horizontal = flat level extrados.
extrados_mode = horizontal
thickness = 0.6
# arch_width is the out-of-plane width b of the analysed arch strip.
arch_width = 0.5
masonry_unit_weight = 24.3
block_count = 20
point_load = 2.6 326 5.2
fill_height = 0.05
fill_unit_weight = 21.76
q_code = 1
fill_spread_angle_deg = 0
friction_coefficient = 0.6
compression_strength = 1500
```

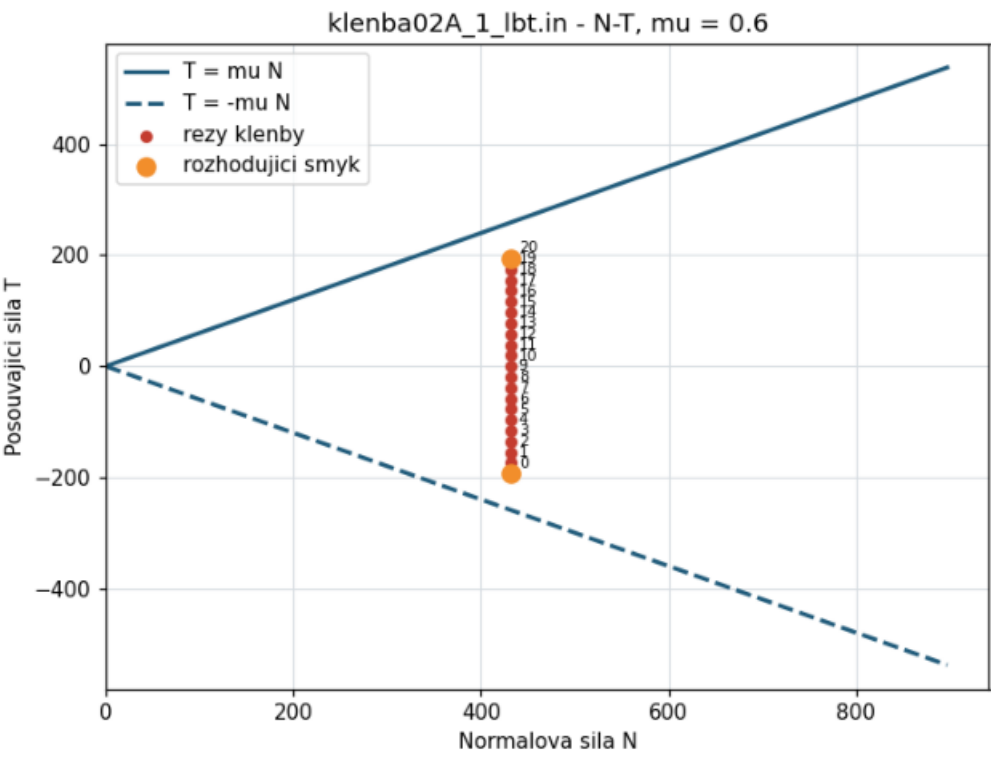
2) Obrazek Klenba



3) Graf M-N



4) Graf N-T



5) Vysledky

rez	xi	yi	xe	ye	N	M	T	e = M/N
0	-0	0	-0	1.15	431.25	-123.98437	-193.61482	-0.2875
1	0.24694569	0.10326891	0.24694569	1.15	431.25	-100.83392	-174.27332	-0.23381778
2	0.49798112	0.19615515	0.49798112	1.15	431.25	-79.550016	-154.91065	-0.18446381
3	0.75266999	0.27849727	0.75266999	1.15	431.25	-60.323512	-135.53731	-0.13988061
4	1.0105696	0.35015215	1.0105696	1.15	431.25	-43.32247	-116.16101	-0.1004579
5	1.2712318	0.41099525	1.2712318	1.15	431.25	-28.692211	-96.786965	-0.066532664
6	1.5342034	0.46092083	1.5342034	1.15	431.25	-16.555399	-77.418228	-0.03838933
7	1.7990275	0.49984212	1.7990275	1.15	431.25	-7.0121484	-58.055991	-0.016260054
8	2.0652436	0.52769145	2.0652436	1.15	431.25	-0.14016043	-38.699956	-0.00032500968
9	2.3323893	0.54442044	2.3323893	1.15	431.25	4.0051638	-19.34869	0.0092873364
10	2.6	0.55	2.6	1.15	431.25	5.390625	-6.1154594e-08	0.0125
11	2.8676107	0.54442044	2.8676107	1.15	431.25	4.0051638	19.34869	0.0092873364
12	3.1347564	0.52769145	3.1347564	1.15	431.25	-0.14016039	38.699956	-0.0003250096
13	3.4009725	0.49984212	3.4009725	1.15	431.25	-7.0121483	58.055991	-0.016260054
14	3.6657966	0.46092083	3.6657966	1.15	431.25	-16.555399	77.418227	-0.03838933
15	3.9287682	0.41099525	3.9287682	1.15	431.25	-28.692211	96.786965	-0.066532664
16	4.1894304	0.35015215	4.1894304	1.15	431.25	-43.32247	116.16101	-0.1004579
17	4.44733	0.27849727	4.44733	1.15	431.25	-60.323512	135.53731	-0.13988061
18	4.7020189	0.19615515	4.7020189	1.15	431.25	-79.550016	154.91065	-0.18446381
19	4.9530543	0.10326892	4.9530543	1.15	431.25	-100.83392	174.27332	-0.23381778
20	5.2	7.0038708e-10	5.2	1.15	431.25	-123.98437	193.61482	-0.2875

6) Report

ArchLBT prepared model
input = D:\Documents\2026\MASARCH2026\python\klenba02A_1_lbt.in

Geometry
span = 5.2
rise = 0.55
thickness D = 0.6
section depth min/max = 0.6 / 1.15
arch width b = 0.5
geom_CODE = 1
extrados mode = horizontal
blocks = 20
joints = 21

Loads
arch self-weight vertical = 49.345471
backfill vertical = 2.828800
variable vertical = 326.000000
total vertical = 378.174271

Lower-bound skeleton
state = permanent + lambda * variable
solution = statically admissible reactions Rx, Ry, M0
constraints = 0 <= N <= fc*b*D_i and abs(M) <= N*(D_i/2 - N/(2*fc*b))
shear constraint = abs(T) <= mu*N
fc = 1500
b = 0.5
Nmax min/max = 450 / 862.5
mu = 0.6
status = optimal
lambda_max = 1.027777
governing joint = 10
governing M-N joints = 0, 10, 20
governing shear joints = 0, 20
governing compression joints = 10
governing tension joints = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20
Rx = 431.250000
Ry = 193.614817
M0 = 123.984375

Prepared arrays
intrados/extrados = 21 x 4
local coordinates = 21 x 4
masonry loads = 21 x 3
backfill loads = 21 x 3
variable loads = 21 x 3

Global equilibrium check
moment origin = [0, 0]
loads sum Fx = 0.000000000e+00

6) Report / pokračovani

```
loads sum Fy = -3.872296344e+02
loads sum M = -1.006797049e+03
left reaction = [4.312499997e+02, 1.936148173e+02], M = 1.239843750e+02
right reaction = [-4.312499997e+02, 1.936148171e+02], M = -1.239843748e+02
residual sum Fx = 0.000000000e+00
residual sum Fy = 0.000000000e+00
residual sum M = 0.000000000e+00
```

7) Generovana pole

```
local coordinates = 21 x 4
[[ 5.75000000e-01 1.57079633e+00 2.52286128e-01 2.06122734e-01]
 [ 5.23365543e-01 1.57079633e+00 2.55295422e-01 1.82937786e-01]
 [ 4.76922424e-01 1.57079633e+00 2.57995104e-01 1.60265969e-01]
 [ 4.35751366e-01 1.57079633e+00 2.60376315e-01 1.38036635e-01]
 [ 3.99923926e-01 1.57079633e+00 2.62431380e-01 1.16183139e-01]
 [ 3.69502373e-01 1.57079633e+00 2.64153780e-01 9.46422007e-02]
 [ 3.44539583e-01 1.57079633e+00 2.65538118e-01 7.33533254e-02]
 [ 3.25078942e-01 1.57079633e+00 2.66580106e-01 5.22582435e-02]
 [ 3.11154274e-01 1.57079633e+00 2.67276538e-01 3.13003883e-02]
 [ 3.02789781e-01 1.57079633e+00 2.67625285e-01 1.04243935e-02]
 [ 3.00000000e-01 1.57079633e+00 2.67625285e-01 -1.04243934e-02]
 [ 3.02789781e-01 1.57079633e+00 2.67276538e-01 -3.13003882e-02]
 [ 3.11154274e-01 1.57079633e+00 2.66580106e-01 -5.22582433e-02]
 [ 3.25078942e-01 1.57079633e+00 2.65538118e-01 -7.33533253e-02]
 [ 3.44539583e-01 1.57079633e+00 2.64153780e-01 -9.46422005e-02]
 [ 3.69502373e-01 1.57079633e+00 2.62431380e-01 -1.16183138e-01]
 [ 3.99923926e-01 1.57079633e+00 2.60376315e-01 -1.38036635e-01]
 [ 4.35751366e-01 1.57079633e+00 2.57995104e-01 -1.60265969e-01]
 [ 4.76922424e-01 1.57079633e+00 2.55295422e-01 -1.82937786e-01]
 [ 5.23365542e-01 1.57079633e+00 2.52286128e-01 -2.06122734e-01]
 [ 5.75000000e-01 1.57079633e+00 1.00000000e+00 4.48965309e-11]]
```

```
masonry loads = 21 x 3
[[0. 6.59105024 0.80106316]
 [0. 6.10191769 0.75404528]
 [0. 5.6484828 0.70848683]
 [0. 5.23714451 0.66567782]
 [0. 4.87361569 0.62681229]
 [0. 4.56287849 0.59296237]
 [0. 4.30914543 0.56505519]
 [0. 4.11582656 0.54385305]
 [0. 3.98550292 0.52993719]
 [0. 3.91990643 0.52369538]
 [0. 3.91990643 0.52531369]
 [0. 3.98550292 0.53477246]
 [0. 4.11582656 0.55184659]
 [0. 4.30914543 0.57611014]
 [0. 4.56287849 0.60694523]
 [0. 4.87361569 0.64355487]
 [0. 5.2371445 0.68497982]
 [0. 5.64848279 0.73011884]
 [0. 6.10191769 0.77752227]
 [0. 6.59105024 0.82656829]
 [0. 0. 0. ]]
```

```
backfill loads = 21 x 3
[[0. 0.26867691 0.0331743 ]
 [0. 0.27312655 0.03428222]
 [0. 0.27710148 0.03528733]
 [0. 0.2805948 0.03618265]
 [0. 0.28360042 0.03696195]
 [0. 0.28611313 0.03761982]
 [0. 0.28812856 0.03815169]
 [0. 0.28964321 0.03855385]
 [0. 0.29065444 0.03882353]
 [0. 0.29116049 0.03895884]
 [0. 0.29116049 0.03895884]
 [0. 0.29065444 0.03882353]
 [0. 0.28964321 0.03855385]
 [0. 0.28812856 0.03815169]
 [0. 0.28611313 0.03761982]
 [0. 0.28360042 0.03696195]
 [0. 0.2805948 0.03618265]
 [0. 0.27710148 0.03528733]
 [0. 0.27312655 0.03428222]
 [0. 0.26867691 0.0331743 ]
 [0. 0. 0. ]]
```

```
variable loads = 21 x 3
[[ 0. 15.48159511 1.91155658]
 [ 0. 15.73799066 1.97539666]
```

7) Generovana pole / pokracovani

[0.	15.96703268	2.03331271]
[0.	16.16832307	2.08490228]
[0.	16.34151197	2.12980685]
[0.	16.48629837	2.16771438]
[0.	16.60243063	2.19836143]
[0.	16.6897069	2.22153504]
[0.	16.74797549	2.23707416]
[0.	16.77713512	2.24487081]
[0.	16.77713512	2.24487081]
[0.	16.74797549	2.23707416]
[0.	16.6897069	2.22153504]
[0.	16.60243063	2.19836143]
[0.	16.48629837	2.16771438]
[0.	16.34151197	2.12980686]
[0.	16.16832307	2.08490228]
[0.	15.96703268	2.03331271]
[0.	15.73799067	1.97539666]
[0.	15.48159511	1.91155658]
[0.	0.	0.]